Lesson plan · for the teacher

Is this the real model?

AI security · Level 3 · Lesson 4

Time
40 min

Materials
Paper and pencils; A shared class phone or computer, if you have one (to list its apps); Board and chalk

Learning goals

- Explain the AI supply chain and how its links get faked.
- Check a file's fingerprint, and say what a signature adds.
- Keep a list of the AI tools the school uses.

Activity: A paper fingerprint

1. Warm-up (5 min): Ask pupils where the apps on their family phone came from: the official store, a link, a file from a friend? Collect answers on the board.
2. Teach (8 min): Explain that apps and AI models are made by others and can be faked or changed. A fingerprint is a short code computed from a file that changes when the file changes; a signature shows who made it.
3. Paper fingerprint (12 min): Each pupil writes a 12-digit number, adds up its digits and writes the total. Partners swap; one secretly changes one digit and hands it back. The owner adds up again: does the total give the change away?
4. Weak spot (7 min): Challenge pairs to change two digits so the total stays the same. Explain that real fingerprints like SHA-256 are built so this is practically impossible.
5. Wrap-up (8 min): With the class, list the apps on a shared class phone or computer and where each came from. Then give the exit ticket.

No shared device? List the apps pupils know from family phones instead, without any personal details. If you have a phone or computer, show one real SHA-256 fingerprint of a file to finish.

Discussion

- Why is “a friend shared it” not a good enough source?
- What can a fingerprint prove, and what can't it?
- What should go on our school's list of tools?

Exit ticket

- Write where one app you use came from, and how you would check that it's genuine.

Try it at school tomorrow
Make a paper fingerprint: pupils write a long number and add up its digits, a partner secretly changes one digit, and the new total gives it away. Then list where the apps on the class's shared phone or computer came from.

Practice questions · for you

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1 **A free AI marking app from an unknown maker went round the staff WhatsApp group, and three colleagues installed it. What now?**

- ◯ Uninstall it, use tools from official sources, and add approved ones to the school's list
- ◯ Keep it if it marks well
- ◯ Ask the person who shared it whether it's safe

2 **Finish fingerprint(data) so it returns the SHA-256 fingerprint of data as 64 hex characters. The program then checks a genuine file (True) and a changed one (False).**

```
import hashlib, binascii

def fingerprint(data):
    # SHA-256 of data, as 64 hex characters
    return ""

good = fingerprint(b"model v1")
print(fingerprint(b"model v1") == good)
print(fingerprint(b"model v2") == good)
```

Write your program:

3 **Put the lines in order to make verify(): it says “use it” only if the file's fingerprint matches the published one. One line doesn't belong.**

Number the lines in order (1, 2, 3…), and draw one arrow → in front of a line for each indent it needs. Cross out any line that doesn’t belong.

- ☐ `h = hashlib.sha256(data).digest()`
- ☐ `if hexed == published:`
- ☐ `if hexed[:4] == published[:4]:`
- ☐ `hexed = binascii.hexlify(h).decode()`
- ☐ `def verify(data, published):`
- ☐ `return "use it"`
- ☐ `return "delete it"`

4 **What should a school's list of AI tools record for each one? Choose all that apply.**

- ☐ The password of each teacher who uses it
- ☐ Its name and version
- ☐ The fingerprint, when the maker publishes one
- ☐ Who approved it
- ☐ Where it was downloaded from

5 **Changing a single byte of a file flips about half of the bits of its SHA-256 fingerprint.**

- ◯ True
- ◯ False

✎ **Which AI apps do you, your colleagues or your pupils use? Do you know where each one came from and who made it?**

Check your answers

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1 **Uninstall it, use tools from official sources, and add approved ones to the school's list**

Nobody knows who made the app or what it does with pupils' work, and whoever shared it doesn't know either.

2

```
import hashlib, binascii

def fingerprint(data):
    h = hashlib.sha256(data).digest()
    return binascii.hexlify(h).decode()

good = fingerprint(b"model v1")
print(fingerprint(b"model v1") == good)
print(fingerprint(b"model v2") == good)
```

sha256 gives a 32-byte digest; hexlify writes it as 64 hex characters.

3 **Top to bottom: 2, 4, X, 3, 1, 5, 6**

```
def verify(data, published):
    h = hashlib.sha256(data).digest()
    hexed = binascii.hexlify(h).decode()
    if hexed == published:
        return "use it"
    return "delete it"
```

Compute the digest, write it as hex, then compare all 64 characters.

4 **Its name and version; Where it was downloaded from; Who approved it; The fingerprint, when the maker publishes one**

Name, version, source, approver and fingerprint let the school check what it runs and react fast when a problem is announced.

5 **True**

That is what makes a fingerprint useful: in the lab, one changed byte flips about 145 of the 256 bits, so even the smallest tampering is caught.